

Exercise 3(a)

Prototype with Familiar and Unfamiliar Navigation Elements

AIM

To develop a mobile application prototype incorporating both familiar and unfamiliar navigation elements and evaluate usability among different user groups using [Proto.io](https://kikic-code.github.io/proto/).

PROTOTYPE LINK: <https://kikic-code.github.io/proto/>

PROTOTYPE OVERVIEW

A mobile application prototype named ParkPilot was created using Proto.io. The prototype simulates a parking assistance system and includes both conventional and gesture-based navigation methods to study user interaction patterns.

SCREEN-WISE DESCRIPTION

1. Intro / Onboarding Screen — “Get Started”

This is the first screen displayed when the application launches. It contains the app title and a “Get Started” instruction that requires the user to swipe upward to continue.

Familiar Navigation Elements

- Minimal layout similar to modern onboarding screens
- Clear instruction indicating progression
- Visual hierarchy guiding user attention

Unfamiliar Navigation Elements

- Swipe-up gesture instead of a button
- No explicit clickable control
- Requires interpretation of gesture-based interaction

This screen tests whether users can recognize motion-based navigation cues.



2. Authentication Screen — Login / Sign-Up

After the onboarding screen, users are presented with options to log in or create a new account. Selecting “Sign Up” allows new users to proceed.

Familiar Navigation Elements

- Standard authentication interface used in most apps

- Clearly labeled buttons for Login and Sign Up
- Tap-based navigation

Unfamiliar Navigation Elements

- Decision-making required for first-time users
- Users unfamiliar with account systems may hesitate

This screen relies primarily on conventional navigation patterns.



3. Account Creation Screen — “Let’s Begin”

This screen simulates account setup. To continue, users must swipe right on the “Let’s Begin” text.

Familiar Navigation Elements

- Linear progression similar to setup processes
- Clear next-step indication

Unfamiliar Navigation Elements

- Swipe-right action instead of a traditional “Next” button
- Gesture-based confirmation may not be immediately obvious

This screen emphasizes exploratory interaction.



4. Home Dashboard Screen

The Home screen acts as the central hub of the application. It provides access to key features such as Map and Profile.

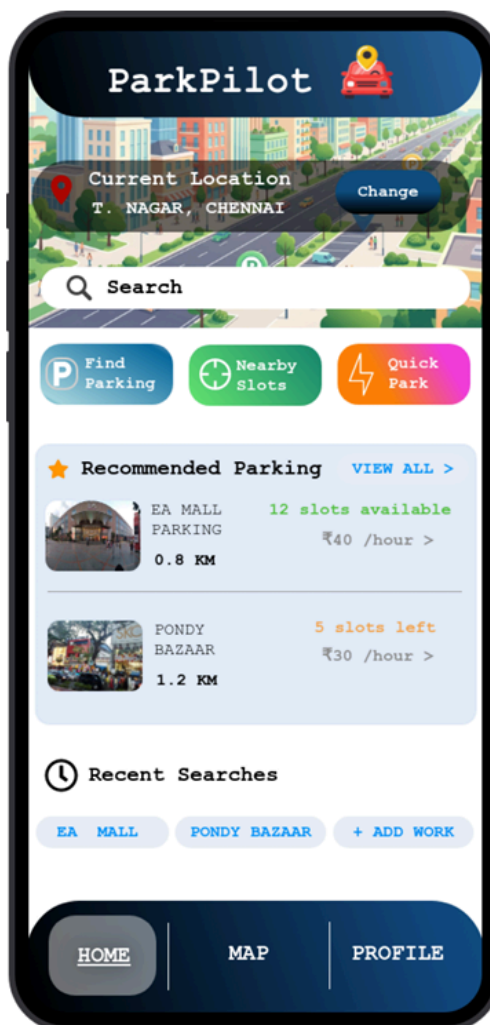
Familiar Navigation Elements

- Dashboard layout common in mobile applications
- Clearly labeled navigation options
- Tap-to-navigate interaction

Unfamiliar Navigation Elements

- First-time users may explore options before understanding functionality
- No tutorial guidance provided

This screen supports orientation within the app.



5. Map Screen — Simulated Navigation View

The Map screen displays a static image representing a navigation route, parking locations, and availability indicators.

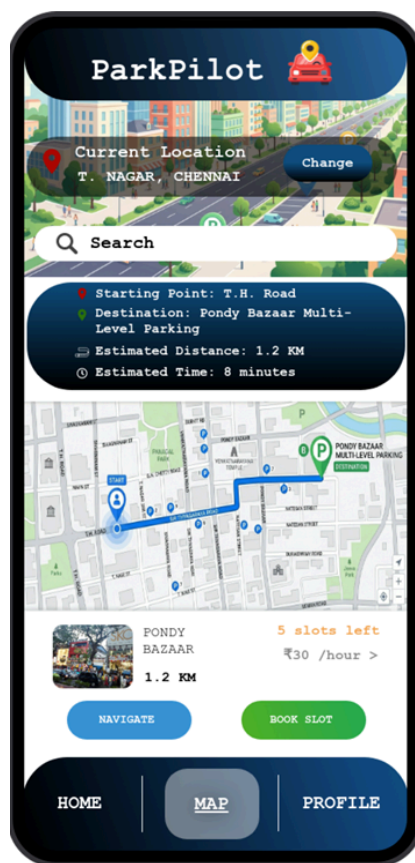
Familiar Navigation Elements

- Visual design resembling popular map applications
- Location pins and route highlighting
- Spatial layout aiding interpretation

Unfamiliar Navigation Elements

- Non-interactive map despite real-world appearance
- Users may expect zooming or panning
- Interpretation of symbols required

This screen focuses on spatial navigation understanding.



6. Profile Screen

The Profile page displays user-related information and provides a Logout option.

Familiar Navigation Elements

- Profile sections are standard across applications
- Logout option placed within account settings
- Structured layout

Unfamiliar Navigation Elements

- Users unfamiliar with app organization may not immediately locate account actions

This screen demonstrates hierarchical navigation.



7. Logout Function — Return to Login Screen

Selecting the Logout option ends the session and redirects the user to the authentication screen.

Familiar Navigation Elements

- Expected behavior in most applications
- Clear session termination

Unfamiliar Navigation Elements

- Loss of access to previous screens may surprise inexperienced users
- Requires understanding of application state change

USABILITY EVALUATION CONSIDERATION

The prototype combines both navigation styles to observe differences in user performance. Familiar elements support efficiency, while unfamiliar elements encourage exploration and learning.

Testing with diverse user groups can reveal how prior experience influences navigation success.

CONCLUSION

The ParkPilot prototype effectively integrates familiar and unfamiliar navigation elements within a single application flow. Using Proto.io, the prototype demonstrates how interface design choices impact usability across different user groups without requiring full system implementation.